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Original Article

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Efficient TurboID-based proximity labelling method for identifying terminal sialic acid glycosylation in living cells

Wu Liu[3,†], Yunfeng Long[2,†], Yongfen Bao[1,†], Yang Li[4], Mengxue Deng[1], Xiuyi Yang[1], He Zhu[5,6,*], and Yanting Su[1,3,*]

[1]School of Basic Medical Sciences, Xianning Medical Colloge, Hubei University of Science and Technology, Xianning 437100, China, [2]School of Pharmacy, Xianning Medical Colloge, Hubei University of Science and Technology, Xianning 437100, China, [3]Hubei Key Laboratory of Diabetes and Angiopathy, Xianning Medical Colloge, Hubei University of Science and Technology, Xianning 437100, China, [4]College of Life Sciences, Wuhan University, Wuhan 430072, China, [5]Hepatic Surgery Center, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan 430030, China, and [6]Hubei Key Laboratory of Hepato-Pancreato-Biliary Diseases, Wuhan 430030, China
[†]These authors contributed equally to this work.
*Correspondence address. Tel: +86-1355436312; E-mail: 2014202040134@whu.edu.cn (Y.S.) / Tel: +86-15527250787; E-mail: whtjzhuhe@163.com (H.Z.)

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Abstract

TurboID, a proximity labelling method based on mutant biotin ligase, is an efficient new technique for recognizing protein-protein interactions and has been successfully applied to living cells. Sialic acid is typically the terminal monosaccharide attached to many glycoproteins and plays many important roles in many biological processes. However, the lack of enrichment methods for terminal sialic acid glycosylation *in vivo* hinders the identification and analysis of this glycosylation. Here, we introduce TurboID to identify terminal sialic acid glycosylation in living cells. *Sp*CBM, the carbohydrate-binding domain of sialidase from *Streptococcus pneumoniae*, is fused with TurboID and overexpressed in HeLa cells. After streptavidin-based purification and detection by mass spectrometry, 31 terminal sialic acid N-glycosylated sites and 1359 putative terminal sialic acid glycosylated proteins are identified, many of which are located in the cytoplasm and nucleus.

Key words TurboID, proximity labelling, glycosylation, sialic acid, *Sp*CBM

Introduction

Recently, enzyme-catalysed proximity labelling (PL) methods have emerged as a new tool to study the interaction between proteins in living cells. In contrast to immunoprecipitation or other affinity purification methods, enzyme-catalysed PL approaches are used in living physiological conditions and reflect real protein–protein interactions [1]. In PL, various labelling enzymes are fused to a target protein and label the interacting proteins by adding a small substrate, such as biotin. APEX2, which is an engineered ascorbate peroxidase from soybean, and BioID, which is a mutant biotin ligase from *Escherichia coli*, are two labelling enzymes that are widely used in PL [2,3]. Nevertheless, all these labelling enzymes have some disadvantages. H_2O_2, a toxic molecule for living cells, is needed for APX2 labelling [4]. The time of BioID labelling is 18–24 h for final proteomic analysis [5]. To avoid these disadvantages, a more efficient and physiological PL tool, namely, TurboID, has been reported [6]. TurboID is a promiscuous labelling variant that is generated from the original BioID and exhibits greater efficiency in producing biotinylation than previous methods. With this new powerful technique, abundant protein-protein interactions were identified in various species, such as mammalian cells, *Drosophila* and yeasts [1,7,8].

Glycosylation is one of the most ubiquitous posttranslational modifications and is linked to many protein functions, such as protein folding, trafficking and binding [9]. Approximately 7000 vertebrate glycan structures that are synthesized by hundreds of glycosyltransferases have been identified within decades [10]. Numerous anomeric linkages, branching and various compositions contribute to the high complexity and structural diversity of glycans [11]. Protein glycosylation is divided into N-linked glycosylation (N-glycan) and O-linked glycosylation (O-glycosylation) according to the nature of linkages and modified amino acids [12]. N-linked glycosylation (N-glycosylation) is linked to asparagine by N-glycosidic bonds and processed in the endoplasmic reticulum followed by the Golgi apparatus. O-glycosylation, also called mucin-

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type O-glycosylation, is covalently linked to serine, threonine, and tyrosine residues in the Golgi apparatus [13]. Therefore, secreted proteins and membrane proteins are regarded as the two main types of glycosylated proteins. However, growing evidence has shown that many proteins in the nucleus and cytoplasm can also be glycosylated [14–17]. For example, nucleolin, which is a kind of nuclear protein, can undergo complex O- and N-glycosylation [14]. Some transcription factors, such as p53 or NFIC, are also reported to be O-glycosylated or N-glycosylated [16,18]. In addition, the latest studies suggest that many glycosyltransferases that are responsible for O-glycosylation are detected in the nucleus. For example, some weak signals of C1GalT1 and ppGalNAc-T3 are found in the nucleus [19,20]. In addition, O-glycosylation can occur in the nucleus independently. These findings indicate that the occurrence of glycosylation in the nucleus or cytoplasm may be a new biosynthetic pathway that is different from the typical pathway, which occurs in the ER and Golgi apparatus. However, how the proteins are glycosylated and which glycosyltransferases are responsible for their modification are still unknown.

Sialic acid is a monosaccharide (N-acetyl neuraminic acid) that frequently occupies the terminal position of glycans. It is reported to largely cap glycans on the cell surface and modulate the interactions between cells along with the cell and extracellular matrix [21]. Many influenza virus infections need to recognize sialic acid on the cell surface via its surface hemagglutinin (HA glycoprotein) [22]. For example, H5N1 and H9N2 can infect the human body because they can bind to the α2,3-linked sialic acid that is enriched on the human epithelial cell membrane [23]. Other virus subtypes that recognize α2,6-linked sialic acid cannot invade the human immune system. Abnormal expression of terminal sialic acid glycosylation results in several pathologies, such as autoimmune disease and tumorigenesis [24,25]. Strikingly, these functions of sialic acid are limited to glycoproteins on the cell surface, and the biological function of sialic acid in the nucleus and cytoplasm is still poorly understood.

At present, *Maackia amurensis* lectin-II (MAL-II, α2,3-linked sialic acid binding lectin) and *Sambucus nigra* agglutinin (SNA, α2,6-linked sialic acid binding lectin) have been widely used to detect sialic acid and uncover the biological functions of sialylated proteins in some diseases [26–29]. Nevertheless, the poor specificity and low affinity of lectin binding to glycans restrict their use on a large scale. From the data of the Consortium for Functional Glycomics (CFG), SNA also recognizes terminal galactose or N-acetylglucosamine glycans with high avidity [30]. MAL-II exhibits binding activity to terminal galactose glycans, as determined by analysis of CFG data. To develop a more powerful tool for identifying sialic acid, a new strategy that mutates the active site of glycosyl hydrolases has been suggested to maintain the high avidity for sialic acid. A carbohydrate-binding domain of sialidase from *Streptococcus pneumoniae*, namely, *Sp*CBM, binds to sialic acid [31]. Multivalent *Sp*CBM has been applied to prevent influenza infection by occluding the interaction between the virus and the terminal sialic acid glycans of cell-surface receptors [32]. Whether *Sp*CBM has higher specificity than other sialic acid binding proteins is still unknown.

In the present study, we synthesized the nucleotide sequences of *Sp*CBM and overexpressed *Sp*CBM proteins using *Escherichia coli*. Purified *Sp*CBM was analyzed by glycan array, and the results showed that *Sp*CBM is a more specific lectin for terminal sialic acid glycans. As it offers an excellent tool, we introduced SpCBM to the enzyme-catalysed proximity labelling technique and detected the terminal sialic acid glycans within cells. In brief, *Sp*CBM was linked to TurboID and overexpressed in HeLa cells. The terminal sialic acid glycosylated proteins were biotinylated by TurboID and purified by a streptavidin-based method. Finally, these proteins were analyzed by mass spectrometry (MS). By this new labelling method, 31 terminal sialic acid N-glycosylated sites and 1359 candidate terminal sialic acid glycosylated proteins were detected. It is worth considering that many glycoproteins are not nonmembrane or nonsecretory proteins.

Materials and Methods

The production of mutant *Sp*CBM

The *SpCBM* gene was synthesized by Sangon Biotech (Shanghai, China) with codon optimization for expression in *Escherichia coli* and then cloned into pGEX-6P-1 with *Bam*HI and *Eco*RI (New England Biolabs, Massachusetts, USA) restriction sites. The prokaryotic expression vector including the *SpCBM* gene was overexpressed in *Escherichia coli* BL21(DE3) expression strains, and cells were cultured overnight at 16°C or 20°C after 1 mM IPTG induction to explore the optimal expression conditions. Finally, 16°C and 12 h were selected as inductive conditions before the cells were harvested by centrifugation at 4200 *g* for 30 min. Cell pellets were resuspended in phosphate-buffered saline (PBS) and lysed by sonication. The cell suspension was purified using GST-tagged Purification Resin (Beyotime, Shanghai, China) according to the manufacturer's instructions.

Biotinylation of *Sp*CBM

A 10-fold molar excess of the biotin reagent purchased from Thermo Fisher Scientific (San Jose, USA) was incubated with 10 mg proteins in PBS for 30 min at room temperature (RT). The free biotin was removed with a Zeba Spin Desalting Column (Thermo Fisher Scientific). The ratio of biotin incorporated into the protein was measured using an HABA assay kit purchased from Thermo Fisher Scientic according to the manufacturer's instructions. Finally, the labelled *Sp*CBM was stored at –20°C.

Glycan array

Glycan array analysis of *Sp*CBM was performed by the Consortium for Functional Glycomics. The printed array consists of 609 glycans in replicates of 6. The glycan array was incubated with 2 concentrations of GST-*Sp*CBM (5 μg/mL and 50 μg/mL).

Hemagglutination assay

Erythrocytes were obtained from four human blood types (A, B, AB and O). Briefly, a series of diluted *Sp*CBM proteins (50 μL, the initial concentration was 0.25 mg/mL) was mixed with 50 μL of erythrocytes. The reactions were performed at room temperature and measured after 1 h.

Cell culture, construction of plasmids and stable cell line generation

HeLa and HEK293T cells were provided by Professor Hui Sun (Wuhan University, Wuhan, China). All cells were maintained in DMEM with 10% fetal bovine serum (FBS; Yeasen, Shanghai, China with 5% CO_2. For western blot analysis, cells were grown in 24-well plates with 500 μL growth medium.

For plasmid construction, the *SpCBM* gene was inserted into engineered pLenti CMV Puro, including the TurboID sequence and

flag tag, which was provided by Professor Hui Sun. To generate lentiviruses, mixed plasmids containing recombinant plasmids and lentiviral packaging plasmids were transfected into HEK293T cells using liposomal transfection reagent (Yeasen) according to the manufacturer's instructions. The lentiviruses were harvested 48 h after transfection.

To generate stable cell lines, HeLa cells were infected with lentiviruses at ~20% confluence. After growing for 24 h, 2 μg/mL puromycin (Yeasen) in growth medium was used to select the positive stable cell lines. Before further analysis was performed, at least 7 days were necessary for selection with puromycin.

Influorescence analysis

The *Sp*CBM-HeLa cells were cultured in 12-chamber culture slides and then fixed with 4% paraformaldehyde for 10 min at 25°C. Then, the cells were permeabilized with 10% Triton X-100 after washing three times in 0.01 M PBS (pH 7.4). The slides were blocked with 3% bovine serum albumin after wash with PBS. The cells were then subject to staining with biotin-SNA or biotin-MAL-II (Vector Laboratories, Burlingame, USA) for 1 h at 25°C, followed by washing with PBS. The cells were then incubated with anti-FLAG antibody (Proteintech, Wuhan, China) for 12 h at 4°C and washed with PBS. Cy3-conjugated AffiniPure goat anti-mouse IgG (H + L) (Proteintech) and FITC-streptavidin (Proteintech) were added to the cell suspension at a ratio of 1:100 (v/v) and incubated for 45 min at 25°C. The cells were then stained with DAPI (5 μg/mL) for 10 min at 25°C. After being washed with PBS, the slides were examined with a Leica SP8 confocal laser scanning microscope (Leica, Wetzlar, Germany).

Biotin labelling with TurboID in living HeLa cells

To label the *Sp*CBM-HeLa cells in 24-well plates, we performed biotin labelling from 10 min to 24 h by adding 500 μM biotin, 1 mM ATP and 5 mM $MgCl_2$ to serum-containing growth medium to explore the optimal labelling time. Then, 150 mm cell plates were applied to the large-scale biotin labelling. Labelling was stopped by transferring the labelled cells on plates to ice, followed by six times wash with PBS.

Protein extraction and western blot analysis

The cells were lysed with ice-cold RIPA buffer (50 mM Tris-HCl pH 8.0, 150 mM NaCl, 0.1% SDS, 0.5% sodium deoxycholate, and 1% Triton X-100) containing 1 × Protease Inhibitor Cocktail (MedChem Express, Monmouth Junction, USA). Cell lysates were clarified by centrifugation at 12,000 g for 10 min at 4°C. The protein concentration of lysates was quantified using a BCA Protein Assay kit (Thermo Fisher Scientific). Then, 20 μg of total protein or proteins which were treated with peptide-N-glycosidase F named PNGase F (New England Biolabs) or sialiase (New England Biolabs) was separated by 10% SDS-PAGE and transferred to PVDF membranes (Millipore, BIllerica, USA). After being blocked with 5% BSA, the biotinylated proteins were identified with HRP-conjugated streptavidin (1:5000; Proteintech). For the western blot analysis of *Sp*CBM, PVDF membrane was incubated with 20 μg/mL biotinylated *Sp*CBM. Immunoblotting was performed using a mouse monoclonal anti-FLAG antibody (1:1000; Proteintech) and anti-GAPDH antibody (1:5000; Proteintech). The membrane was incubated with HRP-conjugated AffiniPure goat anti-mouse IgG (H + L) secondary antibody (1:10,000; Proteintech). Protein detection was performed using the iBright imaging systems (Invitrogen, Carlsbad, USA).

Coimmunoprecipitation

Mock HeLa and stable HeLa cells were harvested and lysed with immunoprecipitation (IP) lysis buffer (25 mM Tris-HCl, 150 mM NaCl, 1 mM EDTA and 1% NP-40) containing 1 × Protease Inhibitor Cocktail for 30 min at 4°C. After centrifugation, the supernatant was incubated with anti-FLAG-coated magnetic beads (Biolinkedin, Shanghai, China) or specific antibodies bound to Protein A/G agarose beads (Biolinkedin) at 4°C for 3 h. All antibodies for immunoprecipitation were purchased from Proteintech. Then, the beads were washed 5 times with IP lysis buffer and boiled with 100 μL of 1 × protein loading buffer at 100°C. Finally, the samples were detected by western blot analysis.

Lectin pull-down assays

To detect the terminal sialic acid on proteins, SNA and MAL-II lectin conjugated agarose beads (Vector Laboratories) were used. Briefly, 2 mg *Sp*CBM-HeLa cell lysates were incubated with SNA or MAL-II conjugated agarose beads overnight at 4°C. Then, the beads were washed 5 times with ice-cold RIPA buffer to eliminate the unbound proteins. Finally, the beads were treated with loading buffer and subject to western blot analysis.

Enrichment of biotinylated proteins

Cells were labelled with biotin for 10 min. To capture biotinylated proteins, the total proteins of mock HeLa cells, which was the control sample, or *Sp*CBM-HeLa cells from two 150-mm culture plates were incubated overnight at 4°C with 200 μL streptavidin-coated magnetic beads (Thermo Fisher Scientific) which had been washed twice with ice-cold RIPA buffer (50 mM Tris-HCl pH 8.0, 150 mM NaCl, 0.1% SDS, 0.5% sodium deoxycholate, and 1% Triton-100) . The magnetic beads were subsequently washed twice with RIPA buffer, once with 1 M KCl, three times with buffer containing 2 M NaCl and 10 mM Tris-HCl, pH 7.4, once with 0.1 M Na_2CO_3, once with 10% SDS, and once with RIPA buffer. Then, the samples were maintained at 50°C for 1 min, followed by 2 times wash with RIPA buffer. Finally, the biotinylated proteins were released from beads by boiling the beads in 100 μL of 1 × protein loading buffer. The efficiency of the enrichment of biotinylated proteins was analysed by western blot analysis and silver staining.

MS sample preparation

Thirty microlitres of enriched proteins and control samples from mock cells that had been denatured were digested by filter-aided sample preparation (FASP) as described previously [33]. Each sample had two replicates. Briefly, the buffer of the sample was replaced by a UA buffer containing 8 M urea and 150 mM Tris-HCl, pH 8.5 in a Millipore filter by sequential centrifugation. Then, the denatured proteins were alkylated with 50 mM iodoacetamide (IAA) in a filter followed by washing three times with UA buffer and three times with 25 mM $NaHCO_3$. The remaining part in the top of the filter was treated with trypsin at 37°C for 18 h. The peptides were collected from the filtrate in the filter by centrifugation and washed with 25 mM $NaHCO_3$ in a new filter. Finally, the peptides were desalted using a C18 Sep Pak cartridge and then lyophilized.

To identify the N-linked glycosylated sites, the peptides were redissolved in 25 mM $NaHCO_3$ solution prepared in $H_2{}^{18}O$ and deglycosylated by PNGase F at 37°C for 3 h. To eliminate the effect

of PNGase F in MS, the naked peptides were collected by filter-aided methods and lyophilized.

Finally, the samples were dissolved in 0.1% formic acid and then analysed by LC-MS/MS in Shanghai Omicsspace Biotech (Shanghai, China). Obitrap Fusion Lumos (Thermo Fisher Scientific) was used to identify the peptides.

MS data analysis

The collected raw data were searched and analysed using Proteome Discoverer 2.4 (Thermo Fisher Scientific) built in Mascot 2.3 employing a human proteome file entitled "human_192901_20201124.fasta" that was downloaded from the UniProt database. Two missed cleavages were allowed during searching. The carbamidomethylation of cysteine was set as a fixed modification, while protein N-terminal acetylation, methionine oxidation and deamidation were set as follows: ^{18}O (NQ) and Asn→Asp (N) were selected as dynamic modifications for searches. The maximum precursor mass tolerance was set to 10 ppm, and product ion tolerance was set to 0.05 Da. Based on these results, the following criterion was established. Proteins were considered glycosylated proteins with the terminal sialic acid for hits in the enriched proteins that exhibited≥2 high-quality peptides and peptide spectrum matches ratio (PSM ratio)≥2. The values of PSM and unique peptides were averaged from two replicates. The PSM ratio was used to exclude nonspecific adsorption and was calculated as the PSM identified in enriched proteins divided by the PSM of the control sample. The criteria were referred to in the paper published in the Journal of Biological Chemistry [20]. The DAVID database (david.ncifcrf.gov) and R software were used to perform GO enrichment analysis and KEGG analysis.

Results

Purification and activity of *Sp*CBM

As shown in Figure 1A, *Sp*CBM is the N-terminal carbohydrate-binding domain (CBM). To obtain enough *Sp*CBM, the *Sp*CBM DNA sequence linked to the C-terminus of the GST tag was transformed

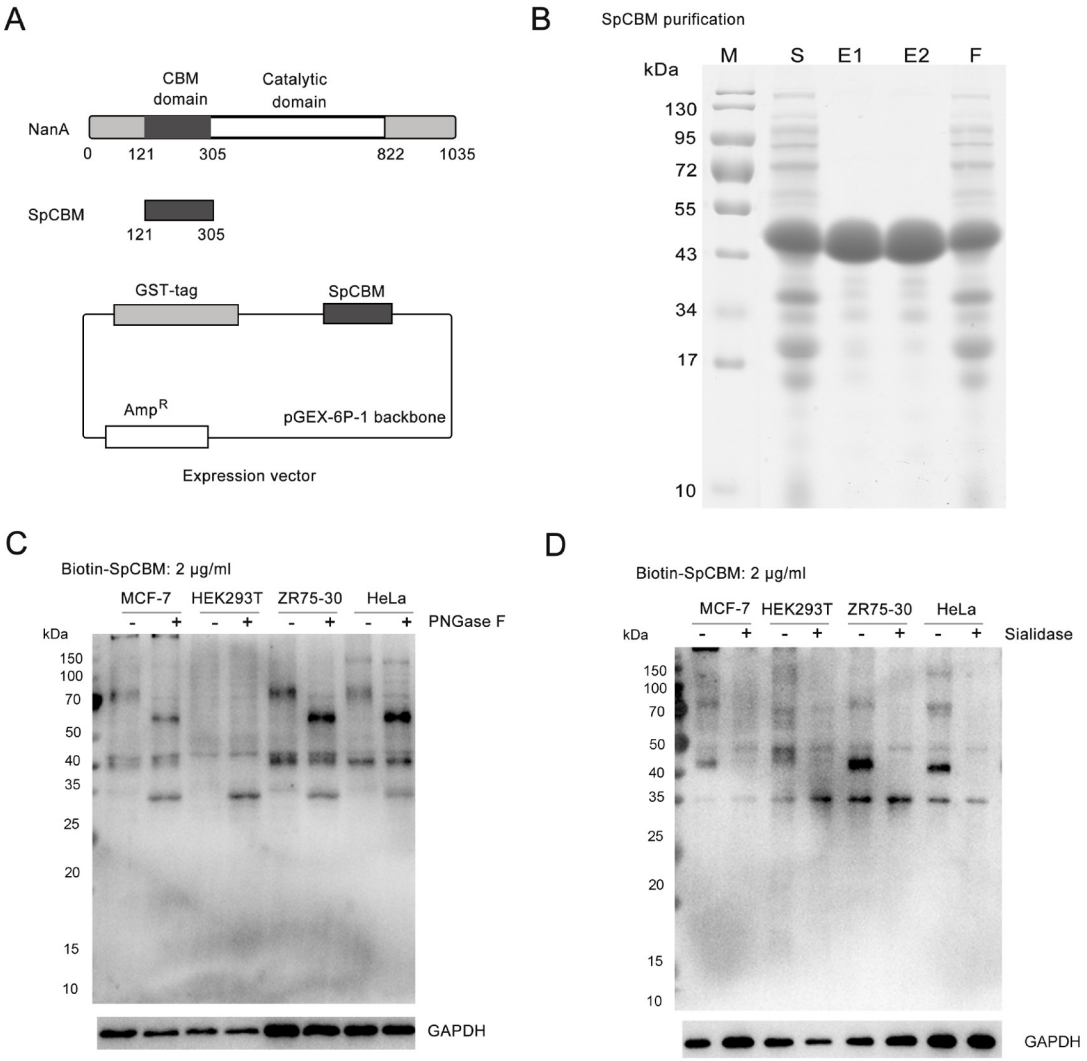


Figure 1. The production and activity of *Sp*CBM (A) The characteristics of *Sp*CBM and detailed strategy of expression for *Sp*CBM in *E. coli*. (B) The purification of *Sp*CBM. S represents the supernatant after ultrasonication. E1 and E2 represent two separated elution samples from the affinity column. F represents the flow through the affinity column. (C) Western blot analysis of *Sp*CBM using an antibody. The cell lysates of different cells were treated with PNGase F and identified with biotin-*Sp*CBM followed by incubation with streptavidin-HRP. (D) Western blot analysis of *Sp*CBM after the sialidase treatment.

into vector pGEX-6P-1 and overexpressed in *E. coli* BL21 (DE3). When the expression was induced at 20°C, more soluble proteins with the predicted molecular weight (45 kDa) were produced (Supplementary Figure S1). The *Sp*CBM protein was purified using GST-tagged Purification Resin and analyzed by SDS-PAGE (Figure 1B). The results showed that the *Sp*CBM protein was highly produced and purified.

To detect the glycan binding activity of *Sp*CBM, we utilized the human cell lysate as the sample because abundant sialylated glycoproteins are present within the cells. To distinguish the type of glycosylation, PNGase F was used to cleave the N-glycans from glycoproteins. Some bands were removed after PNGase F treatment; however, most of them were still retained (Figure 1C). The results indicated that *Sp*CBM might not bind with N-glycosylated proteins. To ensure that *Sp*CBM binds to these proteins depending on sialic acid, the cell lysates were treated with sialidase, which can remove α2,3-, α2,6-, and α2,8-linked sialic acid. Most of the signals identified by *Sp*CBM disappeared after sialidase treatment (Figure 1D). These results suggested that SpCBM recognizes sialic acid glycosylated proteins with high specificity.

Glycan binding profiles of *Sp*CBM

A printed glycan array from the Consortium for Functional Glycomics that consists of 585 glycans was used to analyse the glycan binding profiles of *Sp*CBM. The binding profiles were compared with two frequently used lectins, namely, SNA and MAL-II (Figure 2A). The SNA and MAL-II data were downloaded from the Consortium for Functional Glycomics. Consistent with many previous results, SNA could recognize α2,6-linked sialic acid with high specificity. Nevertheless, SNA showed some other termini, such as galactose, at high concentrations. MAL-II, which is a commonly used lectin that recognizes α2,3-linked sialic acid, showed low specificity in the glycan array results. In our results, *Sp*CBM showed higher specificity for the terminal sialic acid, including the α2,3 and α2,6-linked linkages. The detailed glycan array results obtained from *Sp*CBM are shown in Figure 2B and Supplementary Table S1. *Sp*CBM can recognize many truncated glycans with terminal sialic acid. The top 10 glycans are short glycans shown in Figure 2B. Among these glycans, 5 in 10 are O-GalNAc-type glycans. The top 10 glycans were compared among *Sp*CBM, SNA and MAL-II, and the results are summarized in Table 1. Among the top 10 glycans, SNA tended to recognize N-glycans with terminal α2,6-linked sialic acid and could not identify O-GalNAc-type glycans. MAL-II bound to truncated glycans, including terminal α2,6-linked sialic acid and some terminal galactose glycans, with high affinity. *Sp*CBM showed a higher binding specificity for terminal sialic acid glycans, including some O-glycans and other truncated glycans. Typical N-glycan structures

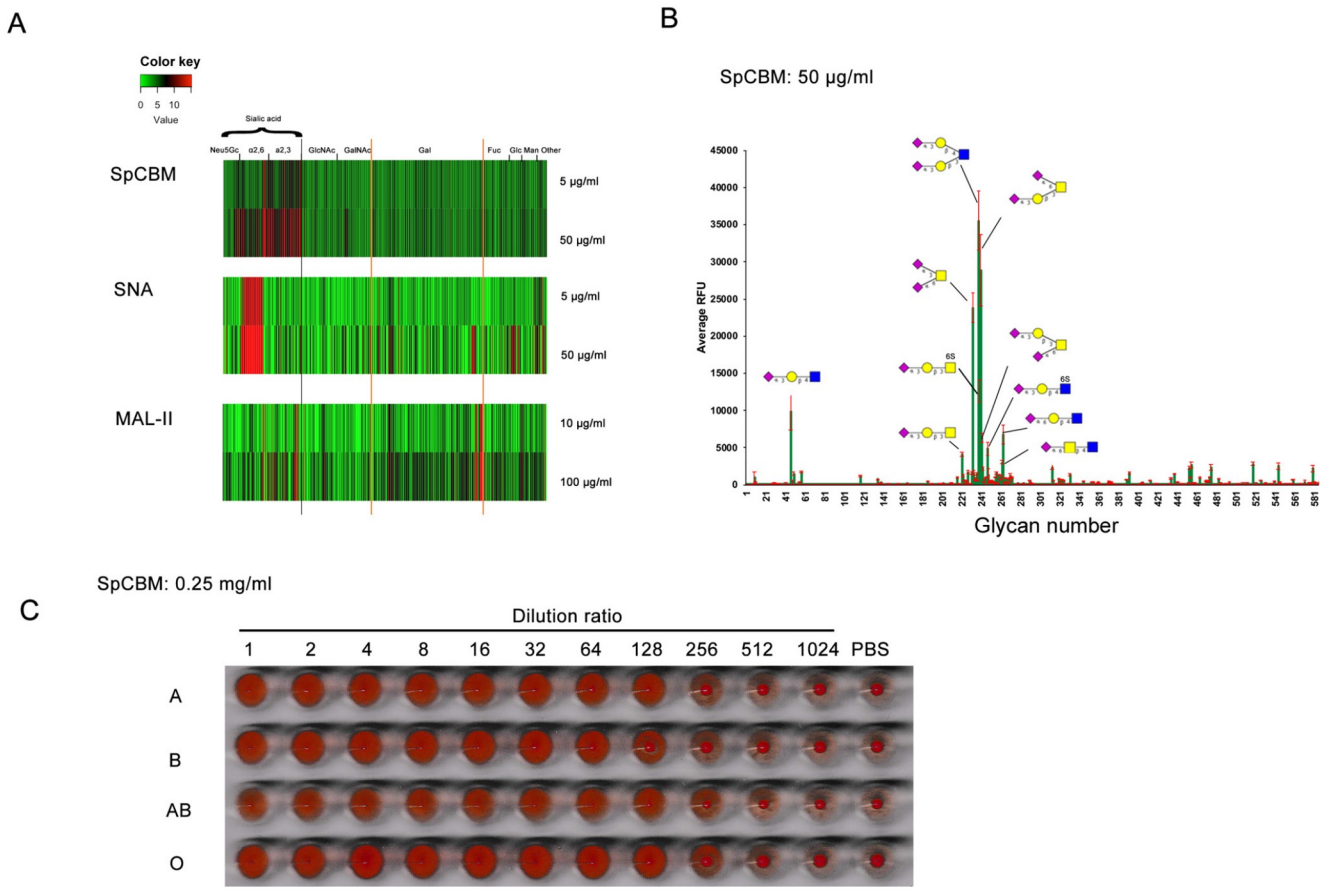
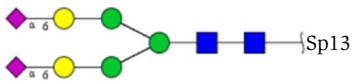
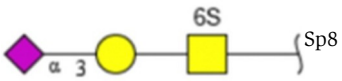
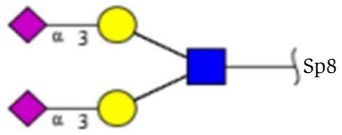
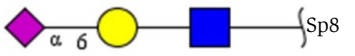
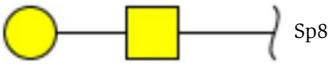
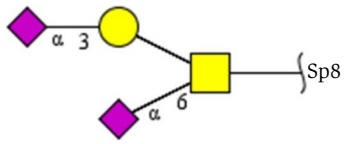
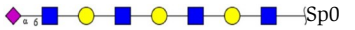
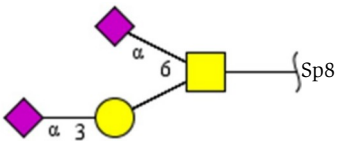
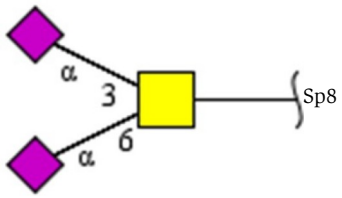
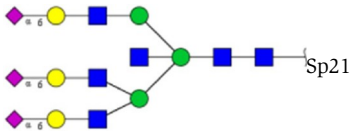
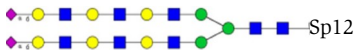
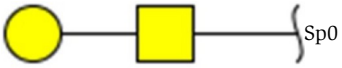
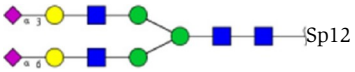
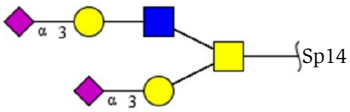
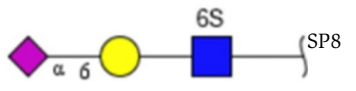
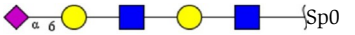
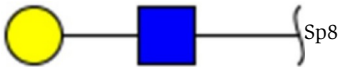
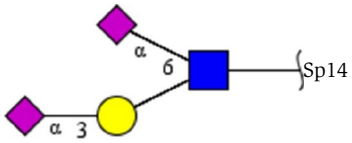
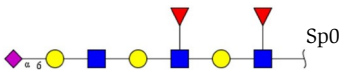
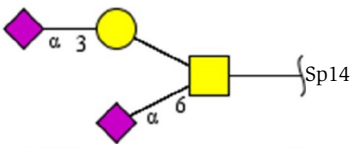
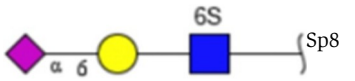
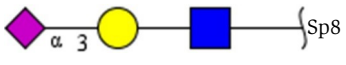
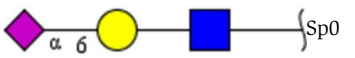
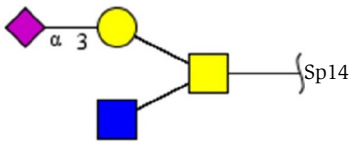


Figure 2. Glycan binding profiles of *Sp*CBM (A) Comparison of glycan array results from CFG with *Sp*CBM, SNA and MAL-II. Red represents strong binding signals, and green represents weak binding or no binding. (B) Top 10 glycans *Sp*CBM binding with high intensity. The purple diamonds represent sialic acid. Blue squares represent GlcNAc. The yellow squares represent GalNAc. The yellow circles represent galactose (Gal). (C) Human erythrocyte agglutination test with *Sp*CBM.

Table 1. The top 10 glycans compared with SNA, MAL-II and SpCBM binding

Ranking	SNA	MAL-II	SpCBM
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			

Purple diamonds represent sialic acid. Blue squares represent GlcNAc. Yellow squares represent GalNAc. Yellow circles represent galactose (Gal). Red triangles represent fucose. Sp is the matrix responsible for linking the glycans. Sp0: -$CH_2CH_2NH_2$. Sp8: -CH_2 $CH_2CH_2NH_2$. Sp12: -Asp. Sp13: -Gly. Sp14: -Thr. Sp21: -$N(CH_3)$-O-$(CH_2)_2$-NH_2.

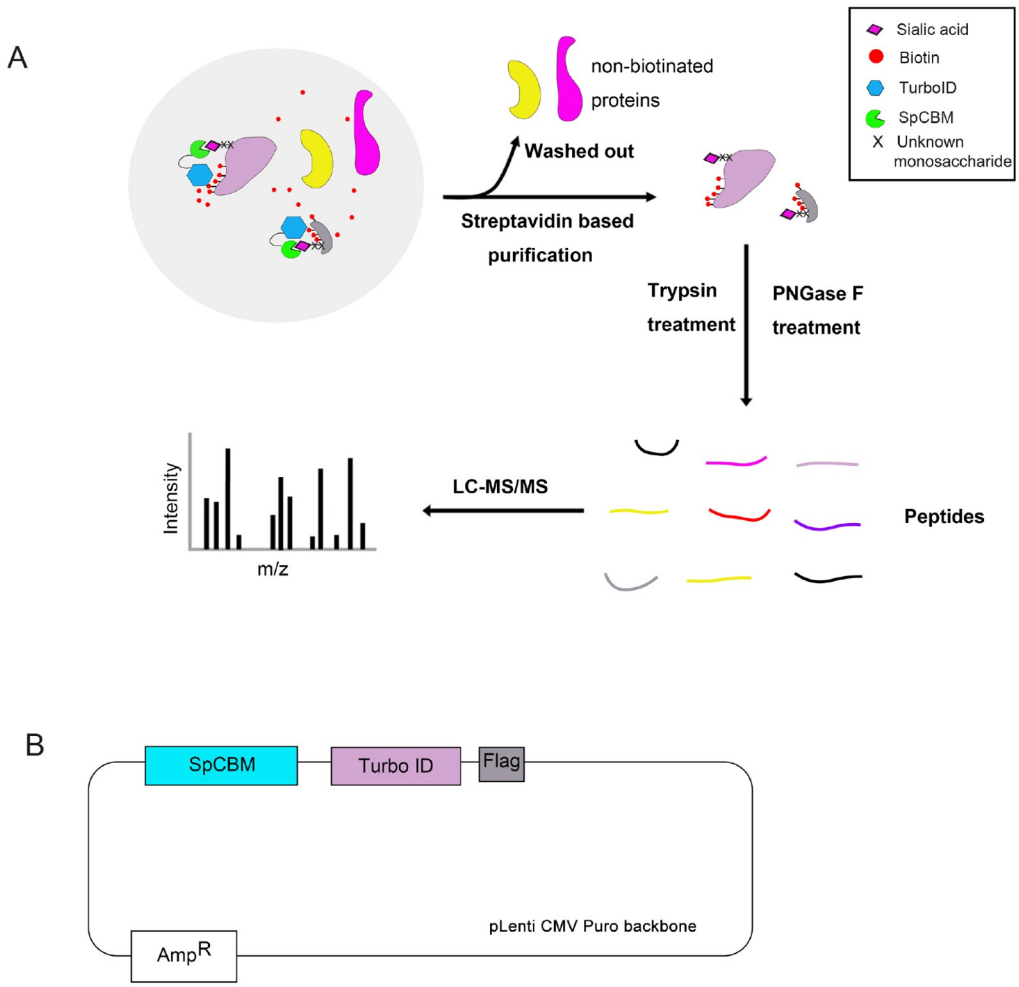


Figure 3. A schedule of TurboID-mediated *Sp*CBM proximity labelling for the terminal sialic acid glycoproteins (A) TurboID was fused to *Sp*CBM and overexpressed in HeLa cells. Offered with biotin, the terminal sialic acid glycoproteins recognized by *Sp*CBM can be labelled with biotin by TurboID. Biotinylated proteins released from the cell lysate were captured by streptavidin-based purification. Enriched proteins were treated with trypsin followed by PNGase F, and peptides were analyzed by MS. (B) Schematic representation of the expression cassettes. *Sp*CBM coupled to TurboID was inserted into pLenti CMV Puro with C-terminal FLAG.

were not identified in the top 10 glycans of *Sp*CBM binding. To verify the results of the glycan array, erythrocytes from human blood (types A, B, AB and O) that contained abundant O-GalNAc-type glycans were used to test the binding of SpCBM (Figure 2C). The results showed that *Sp*CBM could agglutinate all 4 types of erythrocytes, suggesting that *Sp*CBM could recognize the terminal sialic acid O-glycans on the surface of erythrocytes. In conclusion, *Sp*CBM can recognize terminal sialic acid glycans, especially terminal sialic acid O-glycans.

Establishment of protocols for PL in living cells

We exploited a new technique to identify terminal sialic acid glycans by proximity-dependent biotinylation, and the procedure is shown in Figure 3A. We chose *Sp*CBM, a sialic acid-binding domain of sialidase, as a bait protein and coupled SpCBM to the TurboID gene at the pLenti CMV Puro backbone (Figure 3B). By the proximity labelling method, the terminal sialic acid glycosylated proteins that *Sp*CBM can recognize were biotinylated by TurboID and were enriched by streptavidin-based purification. To identify the terminal sialic acid N-glycosylated sites, the N-glycans were removed from glycopeptides in $H2^{18}O$-containing buffer so that the N glycosylated sites were labelled with isotopes and analysed by MS.

Feasibility of *Sp*CBM-TurboID-based proximity labelling for terminal sialic acid

First, the stable cell lines were verified using anti-FLAG antibody (Figure 4A). The results suggested that *Sp*CBM-TurboID was successfully overexpressed in HeLa cells. To determine the optimal labelling time, four time intervals (10 min, 1 h, 12 h and 24 h) were used to label glycosylated proteins after biotin was added to the cell culture medium. Our results showed that biotin labelling by TurboID could be efficient within 10 min, which was consistent with a previous study [6], although extending the time slightly increased the biotinylated signals (Figure 4B).

Sufficient cells were harvested after being labelled and lysed with RIPA buffer. The biotinylated proteins were purified from cell lysates by streptavidin-coated magnetic beads and analysed by silver staining (Figure 4C). The results showed that only biotiny-

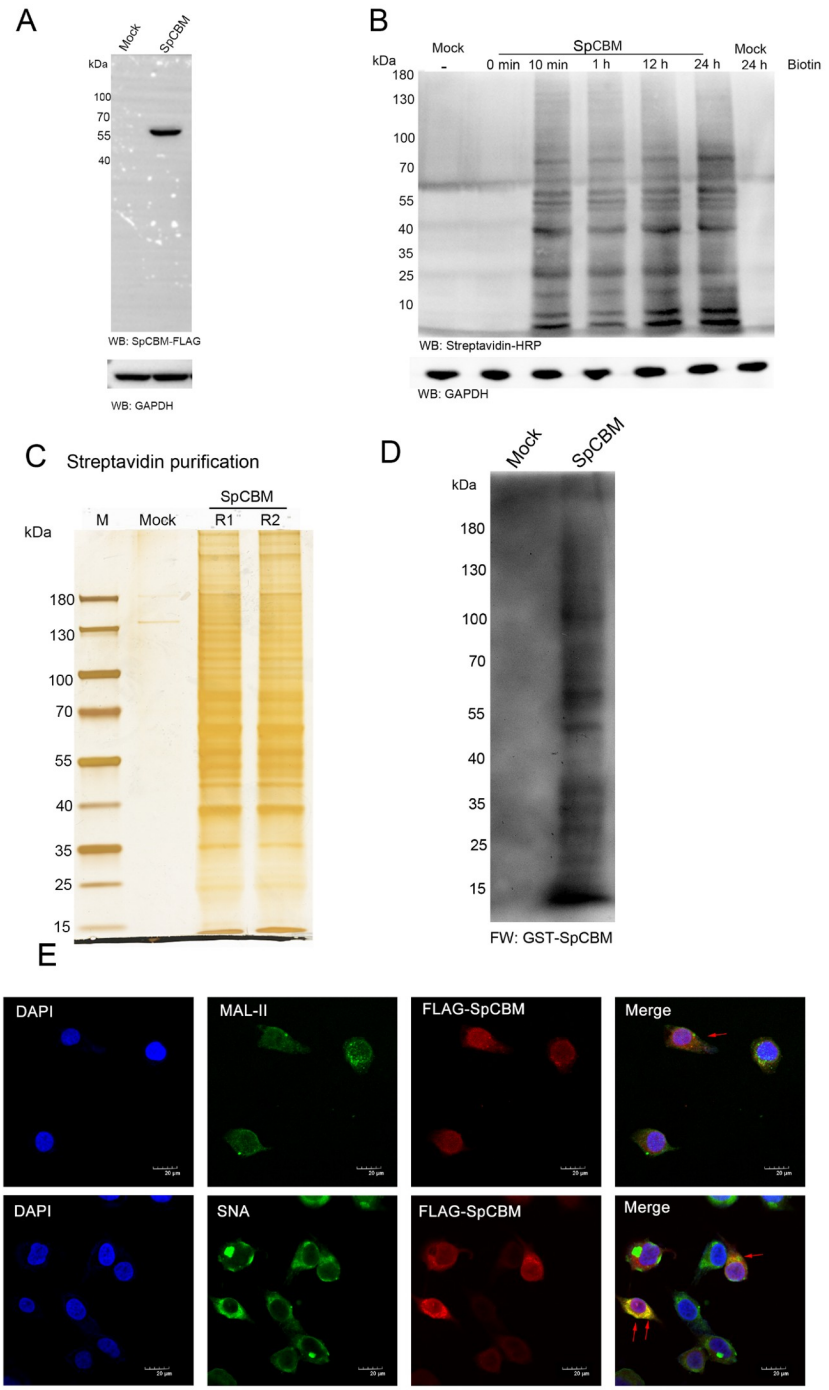


Figure 4. TurboID application in HeLa cells (A) *Sp*CBM coupled to TurboID was overexpressed successfully. The molecular mass of *Sp*CBM linked to TurboID is approximately 55 kDa. (B) Different labelling time (0 min, 10 min, 1 h, 12 h and 24 h) were used. '-' represents the control with no biotin addition. Another control was mock HeLa cell labelling for 24 h. The labelling efficiency was detected using HRP-conjugated streptavidin. (C) The biotinylated proteins were purified with streptavidin-coated beads and analyzed by silver staining. R1 and R2 represent two replicates purified from *Sp*CBM stable cell lysates. (D) The purified biotinylated proteins can be identified by GST-*Sp*CBM-based far western blot analysis (FW). (E) Some colocalized signals between *Sp*CBM and MAL-II or SNA. The red arrows represent strong colocation. Scale bar = 20 μm.

lated proteins were detected in *Sp*CBM stable cell lines, while the lysates from wild-type cell lysates could not be captured by streptavidin. This result indicated that the captured proteins are real biotinylated proteins, namely, the proteins that are bound by *Sp*CBM in cells. In addition, the purified proteins were also determined by incubating GST-*Sp*CBM in western blot analysis (Figure 4D). As expected, strong signals were observed in the purified proteins of *Sp*CBM-overexpressing stable cells compared with those of wild-type cells. These results revealed that the method for identifying terminal sialic acid glycans by *Sp*CBM-based

proximity labelling is feasible in living cells.

Influorescence imaging was also used to detect whether *Sp*CBM-TurboID can label glycosylated proteins with terminal sialic acid in living HeLa cells. SNA and MAL-II lectins were used to identify two terminal sialic acid glycans, namely, α2,6-linked sialic acid and α2,3-linked sialic acid. The results indicated that the *Sp*CBM signal colocalized with the SNA binding or MAL-II signal in some positions (Figure 4E). These data were rational because the glycan binding specificity of *Sp*CBM was not the same as those of SNA and MAL-II, as shown in Figure 2A. In addition, *Sp*CBM was overexpressed in both the nucleus and cytoplasm.

Novel N-glycosylation sites

The glycoproteins purified by the streptavidin-based method were treated with trypsin. The N-linked glycopeptides were deglycosylated in buffer containing $H_2{}^{18}O$. During deglycosylation, asparagine (N) is converted to aspartic acid (D), and the mass difference increases by 0.98 Da compared with that of nonglycosylated peptides under normal conditions, namely, under $H_2{}^{16}O$ conditions. Compared with the deglycosylation in $H_2{}^{16}O$, the mass difference increases by 2 Da in $H_2{}^{18}O$, which is easier for MS identification. By this technique, a total of 31 unique N-glycosylation sites corresponding to 24 glycoproteins were identified. The data that only a few N-glycosylated proteins were identified in the *Sp*CBM recognition sample were consistent with the glycan binding profile of *Sp*CBM. SpCBM binds to truncated glycans, including mainly O-glycans, with high avidity, while long N-glycans were recognized with low avidity. Therefore, SpCBM can capture only a few N-glycosylated proteins with terminal sialic acid. Representative MS/MS spectra derived from several proteins (STIM1, ITGB1 and EIF4B) are shown in Figure 5A–C. Compared with the UniProt database and the results obtained by using the chemical enrichment method [17], we found that only 9 of the 31 sites were reported, and most N-glycosylated sites or glycoproteins were reported for the first time. All N-glycosylated sites are summarized in Table 2. Approximately half of these N-glycosylated sites contained canonical motifs (N-X-S/T motif, X can be any amino acid) (Figure 5D, left panel). However, it must be noted that many N-glycosylated sites do not conform to canonical motifs. Indeed, some N-glycosylated sites with noncanonical motifs have been reported in many samples, such as serum-derived Factor IX [17, 34]. By analyzing the cellular component of these N-glycosylated proteins, we found that many proteins, such as EIF4B, are located in the cytoplasm (which is not an established location for these proteins in contrast to the membrane, especially the cell surface or secreted proteins) (Figure 5D, right panel). It is likely that the N-glycosylated proteins contain noncanonical motifs in cytoplasm.

Functional analysis and validation of the proteome recognized by *Sp*CBM

A total of 1359 glycoproteins were identified in *Sp*CBM proximate proteomes (Supplementary Table S2). Obviously, most of these proteins were not glycosylated with N-glycans. We speculated that most of these glycoproteins were glycosylated with O-glycans because *Sp*CBM binds preferably with O-glycans. Some proteins have been reported to be mucin-type glycoproteins, such as TARS2 and NUFIP2, which are enriched by HILIC methods [35]. In addition, DHX37 and PARP12 we identified were reported to have terminal sialic acid O-glycans [36]. Sialic acid is unstable and easily lost during pretreatment and ionization in MS. It is difficult to identify the terminal sialic acid glycans with MS, while *Sp*CBM coupled to MS may be a good choice for identifying the terminal sialic acid glycosylated proteins.

Subsequently, we categorized the glycoproteins of *Sp*CBM binding by gene ontology (GO) analysis (Figure 6A–C). Cellular components, including cell-cell adherens junction, nucleoplasm, cytoplasm, membrane, nucleolus, nucleus, cytosol, focal adhesion, catalytic step 2 spliceosome and spliceosomal complex, were

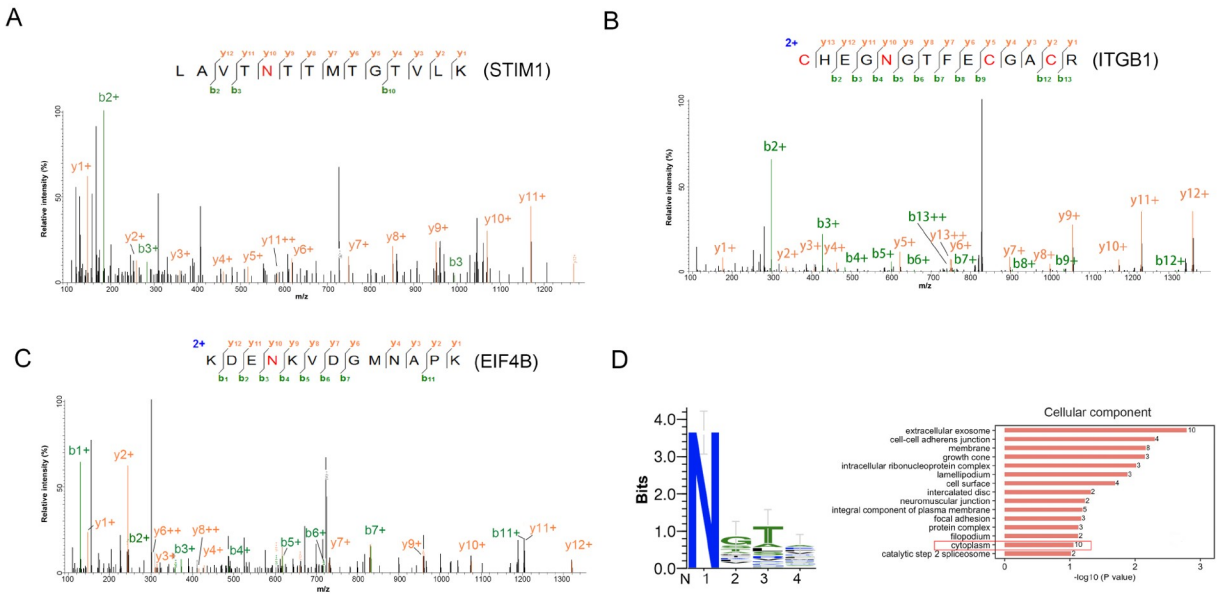


Figure 5. Typical MS/MS spectra and locations of N-glycosylated peptides (A) The tandem mass spectrum of STIM1, residues 167–180, confirms the modification at Asn-171. (B) The tandem mass spectrum of IGTB1, residues 477–490, indicates N-glycosylation at Asn-481. (C) The tandem mass spectrum of EIF4B, residues 530–542, indicates N-glycosylation at Asn-533. (D) The left panel shows the sequence analysis of N-glycosylated sites, and the right panel shows the enriched N-glycosylated proteins of SpCBM based on biological processes.

Table 2. N-glycosylated peptides identified from HeLa cells

Accession	Protein	Peptide	Site	New site/ protein
A0A3G1LBG2	Keratin 1	NGAVTHIKIQNTGDYYDLYGGEK	N48	Y/Y
B4DW52	cDNA FLJ55253, highly similar to actin, cytoplasmic 1	HQGVMVGIVTNWDDMEK	N50	Y/Y
C9JZR2	Catenin delta-1	KNGNGGPGPYVGQAGTATLPR	N187	Y/Y
E7EX17	Eukaryotic translation initiation factor 4B	KDENKVDGMNAPK	N533	Y/Y
F8W914	Reticulon	YSNSALGHVNCTIK	N246	Y/Y
H6VRG2	Cytokeratin-1	NKLNDLEDALQQAK	N445	Y/Y
I3L0A0	HCG2044781	INMNGVNSSNGVVDPR	N320	Y/Y
J3KTA4	DEAD box protein 5	TQNGVYSAANYTNGSFGSNFVSAGIQTSFR	N542	Y/Y
O95297	Myelin protein zero-like protein 1	EIFVANGTQGK	N50	N/N
P05556	Integrin beta-1	CHEGNGTFECGACR	N481	N/N
P05556	Integrin beta-1	KDTCTQECSYFNITK	N669	N/N
P05556	Integrin beta-1	NPCTSEQNCTSPFSYK	N212	N/N
P11717	Cation-independent mannose-6-phosphate receptor	MSVINFECNK	N435	N/N
P11717	Cation-independent mannose-6-phosphate receptor	TNITLVCKPGDLESAPVLR	N581	N/N
P11717	Cation-independent mannose-6-phosphate receptor	DAGVGFPEYQEEDNSTYNFR	N747	N/N
P22626	Heterogeneous nuclear ribonucleoproteins A2/B1	GGGGNFGPGPGSNFR	N226	Y/Y
Q2TTR7	Receptor protein-tyrosine kinase	DSLSINATNIK	N352	Y/Y
Q2TTR7	Receptor protein-tyrosine kinase	TCPAGVMGENNTLVWK	N604	Y/Y
Q2TTR7	Receptor protein-tyrosine kinase	NCTSISGDLHILPVAFR	N361	Y/Y
Q53F48	Heterogeneous nuclear ribonucleoprotein H3 isoform a variant	EIAENALGK	N72	Y/Y
Q5T0F9	Coiled-coil and C2 domain-containing protein 1B	IFSELNSTEMHLIIVR	N697	Y/Y
Q7Z417	Nuclear fragile X mental retardation-interacting protein 2	TGYGELNGNAGEREISLKNLSSDEATNPISR	N98	Y/Y
Q8N4C8	Misshapen-like kinase 1	KGSVVNVNPTNTRAHSETPEIR	N1002	Y/Y
Q9BY44	Eukaryotic translation initiation factor 2A	GSEGLYMVNGPPHFTESTVFPR	N20	Y/Y
Q9NX58	Cell growth-regulating nucleolar protein	.KKGQEADLEAGGEEVPEANGSAGK	N242	Y/Y
Q9Y4D7	Plexin-D1	INLNESMQVVSR	N500	N/N
Q9Y4D7	Plexin-D1	GCVHGNLTFWYMQNPVITAISPR	N1060	N/N
Q9Y520	Protein PRRC2C	SNEVVAVPTNGTVNNVAQEPVNTLGDISGNK	N1463	Y/Y
V5J3L2	STIM1	LAVTNTTMTGTVLK	N171	Y/Y

Red amino acids are the confident N-glycosylated sites "N" represented no. "Y" represented yes.

significantly enriched in GO analysis (Figure 6A). These proteins were related to many molecular functions, including poly(A) RNA binding, cadherin binding involved in cell-cell adhesion, protein binding, RNA binding, nucleotide binding, ATP-dependent RNA helicase activity, ATP binding, mRNA binding, helicase activity and ATP-dependent RNA helicase activity (Figure 6B). Furthermore, several significant enrichments were observed in the biological process of GO analysis, including cell-cell adhesion, mRNA splicing via the spliceosome, mRNA export from the nucleus, cell division, rRNA processing, mRNA processing, viral transcription, RNA splicing, RNA export from the nucleus and sister chromatid cohesion (Figure 6C). By KEGG pathway analysis, the top 10 significant pathways identified from enriched proteins were RNA transport, spliceosome, mRNA surveillance pathway, ribosome biogenesis in eukaryotes, adherens junction, bacterial invasion of epithelial cells, oocyte meiosis, basal transcription factors, endocytosis and mismatch repair (Figure 6D).

To gain confidence in the data from the proteome recognized by *Sp*CBM, an N-glycosylated protein (STIM1) and an enriched protein (SART1) that was not N-glycosylated were selected for verification by coimmunoprecipitation assays. STIM1 is an N-glycosylated protein that we identified, and SART1 located in the nucleus was detected with high abundance in our MS results. As shown in Figure 7A, *Sp*CBM specifically immunoprecipitated STIM1 and SART1. Two widely used lectins for sialic acid, namely, SNA and MAL-II, were also applied to identify the terminal sialic acid of STIM1 and SART1. The results suggested that only a low level of terminal α2,3-linked sialic acid was recognized in STIM1, and high levels of terminal α2,3-linked sialic acid and 2,6-linked sialic acid were found in SART1 (Figure 7B). In our study, we provided a new technique to identify terminal sialic acid glycans, especially O-glycans, which are presently difficult to analyze.

Discussion

In the last few decades, protein glycosylation, which includes N-linked glycosylation and O-linked glycosylation, has been reported to mainly modify secreted proteins and membrane proteins by the secretory pathway. Many proteins in human plasma or serum,

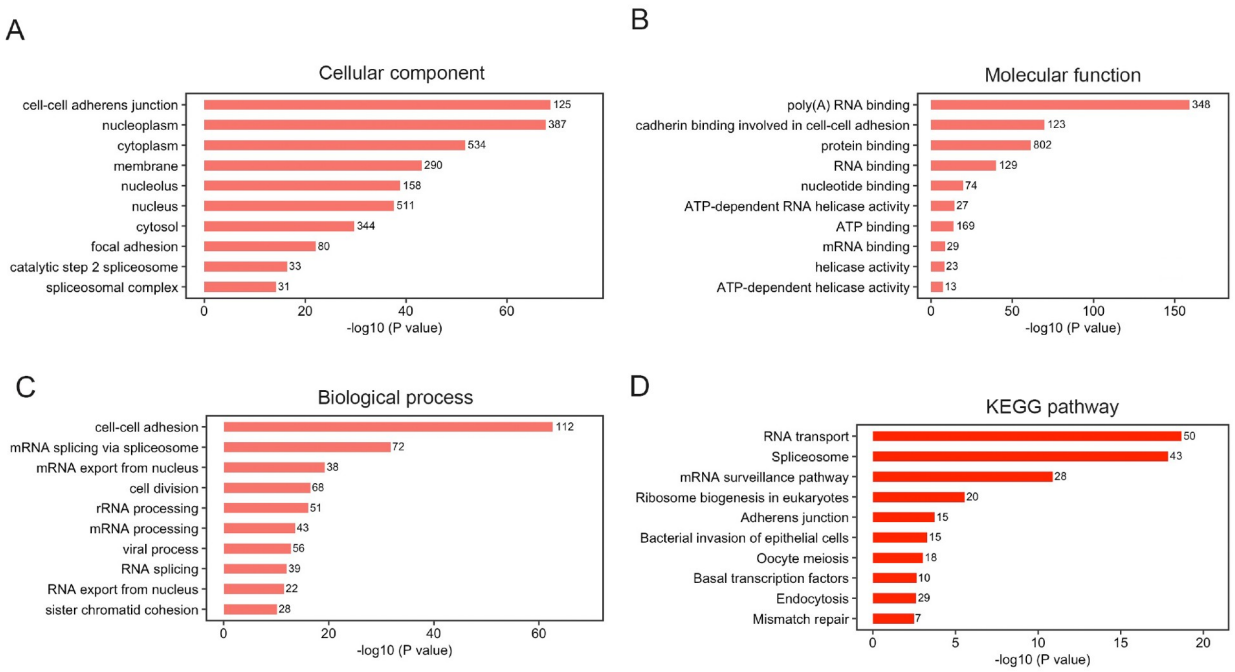


Figure 6. Analysis of *Sp*CBM proximate proteomes (A) Enriched proximate proteins of *Sp*CBM based on cellular components. (B) Enriched proximate proteins of *Sp*CBM based on molecular function. (C) Enriched proximate proteins of *Sp*CBM based on biological process. (D) KEGG analysis of proteins recognized by *Sp*CBM The enrichment of the *P* value is indicated as $-\log_{10}(P)$, and the enrichments of different analyses are ranked by the $-\log_{10}(P)$ values.

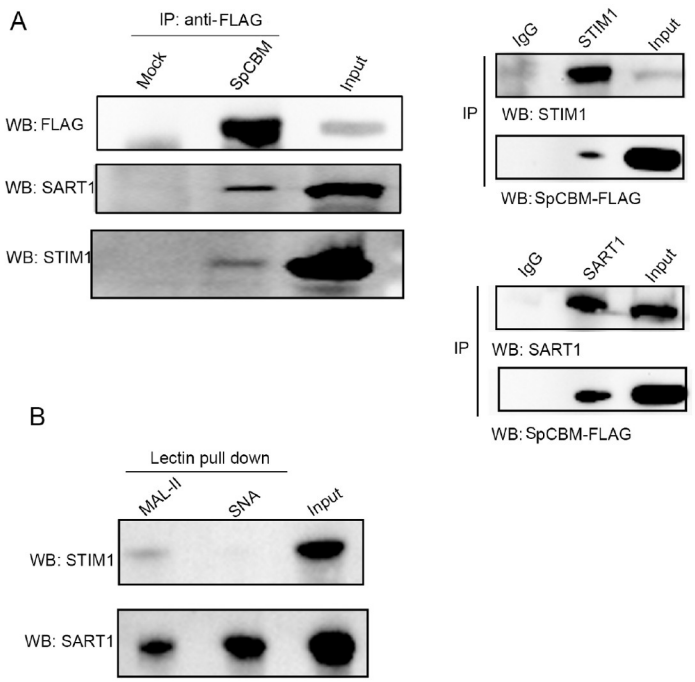


Figure 7. Validation of the interaction between *Sp*CBM and labelled proteins by proximity (A) Co-IP of *Sp*CBM with STIM1 and SART1. The left panel is a co-IP performed with anti-FLAG antibody-coated magnetic beads, and the proteins were identified by anti-STIM1 and anti-SART1 antibodies. The right panel is a co-IP performed with anti-STIM1 and anti-SART1 antibodies coupled to protein A/G beads, and the proteins were identified using anti-FLAG antibody. (B) The terminal sialic acids of STIM1 and SART1 can be captured by SNA or MAL-II pull down assays.

including some disease biomarkers for diagnosis, such as alpha-2-HS-glycoprotein (A2HSG) and antibodies, are highly glycosylated. AHSG contains 2 N-linked glycosylated sites and 2 O-linked glycosylated sites, largely with the terminal sialic acid [37]. Immunoglobulin gamma (IgG) is glycosylated by a conserved N-linked glycosylation site in the heavy chain at Asn297 with complex

types [38]. Except for some glycosyltransferases that initiate the synthesis of N-linked glycans, such as the ALG family within the endoplasmic reticulum, most glycosyltransferases responsible for the synthesis of glycans are membrane proteins in the Golgi apparatus [39,40]. The proteins of the nucleus and cytoplasm can be modified by a single O-linked N-acetylglucosamine (O-GlcNAc) glycosylation, since the O-linked N-acetylglucosaminyltransferase (OGT) responsible for this glycosylation is located in the cell nucleus and cytoplasm [41]. Whether the proteins that are not synthesized by the secret pathway can be glycosylated with complex glycan structures is still a difficult question. However, some glycosyltransferases located in the Golgi apparatus are detected in the cell nucleus. It has been reported that O-linked glycosylation could be synthesized in the cell nucleus and that a GalNAc transferase (polypeptide GalNAc-T3) could also exist in the cell nucleus in addition to the Golgi apparatus [20]. Core 1 β1-3 galactosyltransferase (C1GalT or T-synthase) and its activity for O-linked glycosylation synthesis were detected inside HeLa and HEK-293 cell nuclei [19]. Although no glycosyltransferase for the synthesis of N-linked glycosylation has been described in other cellular locations except the ER and Golgi apparatus, many N-glycosylated proteins within the cytosol have been identified by large-scale glycoproteomic analysis [17,42]. Some proteins in the cell nucleus are also reported to be N-glycosylated [15,16]. Therefore, the glycosylation of proteins can occur in the cell nucleus and cytoplasm, and the identification of these glycosylated proteins will promote the functional study of glycosylation within cells.

The most commonly employed tool to analyse protein glycosylation, including glycosylation site identification and glycan structure determination, is MS. However, it is very challenging to use MS alone to detect protein glycosylation in complex samples without any enrichment [43]. For example, compared to the other monosaccharide, sialic acid is highly unstable and easy to lose during MS analysis. Many derivatization strategies that enhance the stabilities of sialic acid for MS analysis increase the analytical difficulties [44].

The current methods used for the enrichment of glycoproteins or glycopeptides can be divided into biological methods and chemical methods. Lectins, a type of commercially available proteins that bind to glycoproteins, offer biological enrichment approaches. They have a broad specificity for different terminal monosaccharides within the glycan chain, such as concanavalin A (ConA) recognizing mannose, wheat germ agglutinin (WGA) binding to GlcNAc and 2 sialic acid binding lectins: SNA-I and MAL-II [45]. Using SNA or ConA enrichment in combination with MS, the terminal α2,6-linked sialic acid glycopeptides or high mannose-type glycans can be identified in mixed glycoproteins [46]. The enrichment strategy combining antibodies and lectin WGA detected 60 O-GlcNAcylated peptides in mouse embryonic fibroblasts [47]. In addition to lectins, many chemical techniques have been applied to enrich glycopeptides or glycoproteins and identify many glycoproteins, such as hydrazide enrichment and hydrophilic interaction chromatography (HILIC) methods [17,48]. Whether biologically or chemically enriched, all processes occur *in vitro* and not in the physiological conditions of living cells. By using the method established in this study, we designed a new strategy to identify glycoproteins in living cells. By coupling the SpCBM gene with the TurboID gene and overexpressing it in HeLa cells, the terminal sialic acid glycosylated proteins were labelled with biotin by a TurboID-based proximity reaction under living conditions. Finally, the biotinylated glycoproteins were enriched by the streptavidin-based method and identified by MS. Compared with other enrichment methods, such as immunoprecipitation and biochemical fractionation *in vitro*, proximity labelling is more efficient and can reflect living physiological conditions [6]. A total of 1359 putative terminal sialic acid glycoproteins and 31 terminal confident sialic acid N-glycosylated sites were identified by this approach, and most of them were reported for the first time. In addition to heterologous proteins, intrinsic glycosidases and glycosyltransferases are also suitable for labelling different glycosylated proteins by this strategy. Proximity labelling techniques combined with MS will be a powerful tool to identify glycoproteins with different locations on a large scale and will be helpful to expand the research of glycosylation.

Supplementary Data

Supplementary data is available at *Acta Biochimica et Biophysica Sinica* online.

Funding

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Conflict of Interest

The authors declare that they have no conflict of interest.

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